

Dr. Prasenjit Dey

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Summary

Dr. Prasenjit Dey is currently working as an Assistant Professor in the Department of Computer Science and Engineering at the National Institute of Technology (NIT) Rourkela, Odisha, India. He received his B.Tech. from West Bengal University of Technology in 2010, followed by an M.Tech. and Ph.D. in Computer Science and Engineering from NIT Durgapur in 2012 and 2018, respectively. Prior to joining NIT Rourkela, he held positions at Intel as a Hardware Graphics Designer and served as an Assistant Professor at Cooch Behar Government Engineering College.

His research interests span artificial neural networks, machine learning, pattern recognition, deep learning, IoT system design, data visualization, and feature engineering. Dr. Dey has successfully completed a government-funded project and is actively involved in several others. He has published extensively, with more than 30 research papers in reputed international journals such as IEEE Transactions on Systems, Man, and Cybernetics, IEEE Transactions on Artificial Intelligence, IEEE Internet of Things Journal, Pattern Recognition, Biomedical Signal Processing and Control, and Information Sciences, as well as numerous conference proceedings and edited book chapters.

In addition to his strong research profile, Dr. Dey brings exceptional skills in program development, student mentoring, interdisciplinary collaboration, and effective scientific communication.

Brief Synopsis of Research

Dr. Prasenjit Dey's research is rooted in the development of intelligent, interpretable, and robust machine learning systems with applications in real-world domains such as healthcare, surveillance, and human behavior analysis. His work spans deep learning, neural network regularization, explainable AI, and vision-language models, with a particular emphasis on enhancing generalization, stability, and transparency in AI-based decision systems. He has contributed to the design of novel architectures for micro-expression recognition, underwater image enhancement, cognitive load detection using fNIRS, and anomaly detection in dynamic environments.

A significant part of his research focuses on bridging the gap between theoretical modeling and deployable solutions—leveraging transformer-based architectures (e.g., ViTs, VLMs), attention mechanisms, and interpretability tools like Grad-CAM. His publications in high-impact journals such as IEEE Transactions on Artificial Intelligence, Pattern Recognition, and Scientific Reports reflect a strong commitment to both algorithmic innovation and societal relevance. He also explores optimization techniques, feature engineering, and low-

power AI for edge devices, with current involvement in multi-institutional, government-funded projects focused on real-time AI applications.

Experience

➤ Academic Experience

Assistant Professor at National Institute of Technology Rourkela since July 2023

➤ Subject Taught:

S. No.	Courses	No. of times taught	UG / PG	Work Place
1.	Discrete Structures	1	UG	National Institute of Technology Rourkela
2.	Database Management Systems	1	UG	National Institute of Technology Rourkela
3.	Data Science	2	UG	National Institute of Technology Rourkela
4.	Introduction to AI and ML	1	UG	National Institute of Technology Rourkela
5	Software Engineering	2	UG	National Institute of Technology Rourkela

Assistant Professor at Coochbehar Government Engineering College from June 2018 to July 2023

➤ Subject Taught:

S. No.	Courses	No. of times taught	UG / PG	Work Place
1.	Computer Organization	5	UG	Coochbehar Government Engineering. College
2	Computer Architecture	4	UG	Coochbehar Government Engineering. College
3	Multimedia	2	UG	Coochbehar Government Engineering. College
4	Operating System	3	UG	Coochbehar Government Engineering. College
5	Machine Learning	1	UG	Coochbehar Government Engineering. College
6	Big Data	1	UG	Coochbehar Government Engineering. College
7	Cryptography & Network Security	1	UG	Coochbehar Government Engineering. College
8	Object Oriented Programming	1	UG	Coochbehar Government Engineering. College

➤ Research Experience

Research Scholar at NIT Durgapur from December 2012 to June 2018

Main duties performed: Research activities and taking theory and practical classes

➤ Industry Experience

1. Hardware Graphics Engineer at INTEL from February 2017 July till June 2018

Main duties performed: 3-D graphics and media designing

2. **Associate Innovator at NIVIO from July 2012 July till November 2012**

Main duties performed: Web service maintenance

Sponsored Research & Development projects

S. No.	Title	PI and Co-PI	Budget	Under the Scheme	Project Duration	Year of Allocation
1.	Design and Implementation of Real-Time Object Detection Based System for Eliminating Injury and Death of Elephants due to Forest Railway Track Accidents	PI: Aritra Acharyya, Co-PI: Prasenjit Dey	11.99Lakh	Department Of Science & Technology, Govt. of West Bengal (WBDST)	3 Years	2020
2.	Prototyping of ML based Solar and Wind Forecasting Approach and intelligent Power Management Scheme for Hybrid Storages in On-board Naval Microgrid	PI: Arnab Ghosh, Co-PI: Provat Kumar Ray Prasenjit Dey	42.78Lakh	NAVAL RESEARCH BOARD (NRB), DRDO	3 Years	22/4/2024

Areas of Expertise

- Machine Learning and Deep Learning
- Computer Vision and Image Processing
- Sensor Networks and Real-Time Systems
- Neural Network Robustness and Regularization
- Intelligent Embedded and IoT Systems

Research Publications

Journals:

S. No.	Author(s)	Title	Name of Journal	Volume	Page	Year
1.	P Sarkar, Sourav De, Prasenjit Dey , Sandeep Gurung	Advanced YOLOv4 for real-time underwater object detection: An application-oriented approach	Applied Soft Computing DOI: https://doi.org/10.1016/j.asoc.2025.113837	-	113837	2025
2.	Ankita Jain, Dhananjoy Bhakta, Prasenjit Dey	AuLiTENet: A New Lightweight and Time-Efficient CNN Architecture for Micro-expression Classification Leveraging Data Augmentation	Sadhana https://doi.org/10.1007/s12046-025-02781-3	50	143	2025

		DOI:				
3.	Ankita Jain, Dhananjay Bhakta, Prasenjit Dey	Two-tiered Spatio-temporal Feature Extraction for Micro-expression Classification	Journal of Visual Communication and Image Representation DOI: https://doi.org/10.1016/j.jvcir.2025.104436	109	104436	2025
4.	P Sarkar, Sourav De, Sandeep Gurung Prasenjit Dey,	UICE-MIRNet guided image enhancement for underwater object detection	Scientific Reports DOI: https://doi.org/10.1038/s41598-024-73243-9	14	22448	2024
5.	Rahul Mondal, Tandra Pal, Prasenjit Dey	Discriminative Regularized Input Manifold for multilayer perceptron	Pattern Recognition, DOI: https://doi.org/10.1016/j.patcog.2024.110421	151	110421	2024
6.	Prasenjit Dey, Dhananjay Bhakta	A New Random Forest and Support Vector Machine-based Intrusion Detection Model in Networks	National Academy Science Letters DOI: https://doi.org/10.1007/s40009-023-01223-0	46	471-477	2023
7.	Ashok Pradhan, Bhaskara Santhosh Egala, Prasenjit Dey, Venkataramana; Badarla, Saraju Mohanty	Fortified-Chain2.0: Intelligent Blockchain for Decentralized Smart Healthcare System	IEEE Internet of Things Journal DoI: https://doi.org/10.1109/JIOT.2023.3247452	10	12308--12321	2023
8.	Rahul Mondal, Tandra Pal and Prasenjit Dey	A Hybrid Regularized Multilayer Perceptron for Input Noise Immunity	IEEE Transactions on Artificial Intelligence DOI: https://doi.org/10.1109/TAI.2022.3225124	5	115-126	2023
9.	Karmakar, Subashis, Supreeti Kamilya, Prasenjit Dey, Parag K. Guhathakurta, Mamata Dalui, Tushar Kanti Bera, Suman Halder, Chiranjib Koley, Tandra Pal, and	Real time detection of cognitive load using fNIRS: A deep learning approach	Biomedical Signal Processing and Control https://doi.org/10.1016/j.bspc.2022.104227	80	104227	2023

	Anupam Basu					
10	Aritra Acharyya, Prasenjit Dey , Abir Datta, and Sukanya Bose	Analytical Modelling of Video Surveillance System Along Forest Railway Tracks	IETE Journal of Education https://doi.org/10.1080/09747338.2020.1798821	61	55-63	2020
11	Prasenjit Dey , Madhumita Gopal, Payal Pradhan and Tandra Pal	On Robustness of Radial Basis Function Network with Input Perturbation	Neural Computing and Applications, Springer https://doi.org/10.1007/s00521-017-3086-5	31	523-537	2019
12	Prasenjit Dey , Kaustuv Nag, Tandra Pal and Nikhil R. Pal	Regularizing Multilayer Perceptron for Robustness	IEEE Transactions on Systems, Man, and Cybernetics: Systems DOI: https://doi.org/10.1109/TSMC.2017.2664143	48	1255-1266	2017
13	Prasenjit Dey and Tandra Pal	On Improving Generalization of Classifiers Using Feature Extraction	International Journal of Control Theory and Applications	10	29-38	2017
14	Prasenjit Dey and Tandra Pal	A new feature extraction technique for classifiers using self-organising map	International Journal of Convergence Computing https://www.inderscienceonline.com/doi/abs/10.1504/IJCONVC.2016.090079	2	208-219	2016

Conferences:

S. No.	Author(s)	Title	Name of Conference	Page	Year
1.	Prasenjit Dey and Tandra Pal	Extended Self Organizing Map with Probabilistic Neural Network for Pattern Classification Problems	International Conference on Frontiers of Intelligent Computing: Theory and Applications	167-174	2015
2.	Prasenjit Dey , Abhijit Ghosh, and Tandra Pal	Regularized Stacked Auto-encoder based Pre-training for Generalization of Multi-layer Perceptron	International Conference on the Theory and Practice of Natural Computing (TPNC)	232-242	2017
3	Sourabrata Mukherjee, Prasenjit Dey , Tandra Pal	GFRP cutting parameters optimization using different MOEAs	International Conference on Intelligent Computing, Instrumentation and Control Technologies	654-659	2017

			(ICICICT)		
4	Subhradip Aon, Ashis Sau, Prasenjit Dey , Tandra Pal	IMOGA/SOM: An Intelligent Multi-objective Genetic Algorithm Using Self Organizing Map	International Work-Conference on Artificial Neural Networks	40-51	2017
5	Rahul Mondal, Prasenjit Dey , Gautam Sharma and Tandra Pal	Regularizing Multilayer Perceptron for Generalization Using KL-Divergence	International Conference on Computer Science, Engineering and Applications (ICCSEA 2020)	1-6	2020
6	Sourabrata Mukherjee and Prasenjit Dey	Deep Learning-Based Weight Initialization on Multi-layer Perceptron for Image Recognition	Doctoral Symposium on Intelligence Enabled Research (Springer Nature Singapore)	197-206	2022
7	Prasenjit Dey , Sudip Kumar Adhikari, Arnab Gain, and Suman Koner	Quality Analysis of the Ganges River Water Utilizing Machine Learning Technologies	Doctoral Symposium on Intelligence Enabled Research (Springer Nature Singapore)	11-20	2022
8	Sukanya Bose, Swarnava Ghosh, Subhadeep Dhang, Rohan Karmakar, Prasenjit Dey , and Aritra Acharyya	IoT-Based Smart Internet Protocol Camera for Examination Monitoring	Doctoral Symposium on Intelligence Enabled Research (Springer Nature Singapore)	249-260	2022
9	Sukanya Bose, Swarnava Ghosh, Subhadeep Dhang, Prasenjit Dey , and Aritra Acharyya	Design and Realization of an IoT-Based System for Real-Time Monitoring of Embryo Culture Conditions in IVF CO2 Incubators	Doctoral Symposium on Intelligence Enabled Research (Springer Nature Singapore)	161-172	2022
10	Sudipta Hazra, Susneha Ghosal, Arindam Mondal, Prasenjit Dey	Forecasting of Rainfall in Subdivisions of India Using Machine Learning	Doctoral Symposium on Intelligence Enabled Research	205-214	2024

Edited Books:

S. No.	Title	Author's Name	Publisher	Year of Publication
1.	Internet of Things-Based Machine Learning in Healthcare Technology and Applications	Prasenjit Dey , Sudip Kumar Adhikari, Sourav De, Indrajit Kar	CRC Press DoI: https://doi.org/10.1201/9781003391456	2025
2.	Real-World Applications and Implementations of IoT	Aritra Acharyya, Prasenjit Dey , Sujit Biswas	Elsevier DoI: https://doi.org/10.1007/978-981-97-8627-5	2024

Book Chapters:

S. No.	Title	Author's Name	Publisher	Year of Publication
3.	Applications of Internet	Prasenjit Dey ,	CRC Press	2024

	of Things and Machine Learning Technologies in Healthcare	SK Adhikari, S De, R Banerjee	DoI: https://doi.org/10.1201/9781003391456	
4.	Medical image analysis using swarm intelligence: A survey	SK Adhikari, Prasenjit Dey , S De, S Paul	Elsevier DoI: https://doi.org/10.1016/B978-0-443-15533-8.00012-6	2024
5.	A Survey on Energy-Efficient Routing in Wireless Sensor Networks Using Machine Learning Algorithms	Prasenjit Dey and Arnab Gain	Novel Research and Development Approaches in Heterogeneous Systems and Algorithms, IGI Global DOI: 10.4018/978-1-6684-7524-9.ch015	2023
6.	Adaptive Position-Based Crossover in the Genetic Algorithm for Data Clustering	Arnab Gain and Prasenjit Dey	Recent Advances in Hybrid Metaheuristics for Data Clustering, Wiley https://doi.org/10.1002/9781119551621.ch3	2020

Skills

Technical: Cryptographic techniques, Operating system, Data structure, Computer organization, Microprocessor (8085), Database management system, Automata & compiler design, Computer graphics

Languages: Python, R, MATLAB, C, C++, Java, Ruby, Pearl.

Tools: MATLAB, SPSS, R studio, Visual Studio, SVN, Perforce, Github, Android Studio.

Domain: Image Processing, Computer Vision, Machine Learning, Statistical Analysis, Data Analysis, Data Science, Natural Language Processing.

Interpersonal: Punctual, well organized and ability to motivate co-workers

Others: Organized various events as the head of cultural committee at NIT Durgapur, Played cricket and football.

Achievements

- Qualified in GATE 2010
- Qualified in NET 2012
- 2nd Runner-Up in the CMPDI Hackathon on Research and Development, 2025

Ph.D thesis title, Institute/Organization/University, Year of Award.

Title: Designing Neural Network Algorithms for Better Performance Analysis
 University: National Institute of Technology, Durgapur
 Year: 2018

Declaration

I do hereby declare that all the information furnished above is true to the best of my knowledge.

DATE: _____

SIGNATURE: _____

(Prasenjit Dey)



Prasenjit Dey ✎

National Institute of Technology Rourkela

Verified email at nitrkl.ac.in

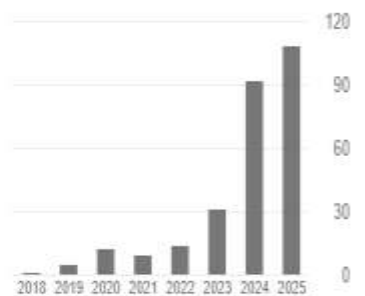
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	Tandra Pal Professor, Department of Compu...	>
	Nikhil R. Pal Indian Statistical Institute	>
	Kaustuv Nag Indian Institute of Information Te...	>

☐ TITLE	CITED BY	YEAR
☐ Fortified-chain 2.0: Intelligent blockchain for decentralized smart healthcare system BS Egala, AK Pradhan, P Dey, V Badarlia, SP Mohanty IEEE Internet of Things Journal 10 (14), 12308-12321	56	2023
☐ Real time detection of cognitive load using fNIRS: A deep learning approach S Karmakar, S Kamliya, P Dey, PK Guhathakurta, M Dalui, TK Bera, ... Biomedical Signal Processing and Control 80, 104227	48	2023
☐ Regularizing multilayer perceptron for robustness P Dey, K Nag, T Pal, NR Pal IEEE Transactions on Systems, Man, and Cybernetics: Systems 48 (8), 1255-1266	47	2017
☐ On robustness of radial basis function network with input perturbation P Dey, M Gopal, P Pradhan, T Pal Neural Computing and Applications 31 (2), 523-537	19	2019
☐ A new random forest and support vector machine-based intrusion detection model in networks P Dey, D Bhakta National Academy Science Letters 46 (5), 471-477	15	2023
☐ UICE-MIRNet guided image enhancement for underwater object detection P Sarkar, S De, S Gurung, P Dey Scientific Reports 14 (1), 22448	14	2024
☐ Forecasting of Rainfall in Subdivisions of India Using Machine Learning S Hazra, S Ghosal, A Mondal, P Dey Doctoral Symposium on Intelligence Enabled Research, 205-214	14	2023
☐ Future Prospects of Agriculture Using IoT and Machine Learning S Hazra, A Mondal, P Dey, S Prabhakar, AK Jha, N Rakshit 2024 IEEE International Conference on Smart Power Control and Renewable ...	10	2024
☐ A hybrid regularized multilayer perceptron for input noise immunity R Mondal, T Pal, P Dey IEEE Transactions on Artificial Intelligence 5 (1), 115-126	8	2023
☐ Adaptive position-based crossover in the genetic algorithm for data clustering A Gain, P Dey Recent Advances in Hybrid Metaheuristics for Data Clustering, 39-59	8	2020